CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

JOHN SILVEIRA DAIRY #2 (HEIFERS)

29687 FENTEM RD

GUSTINE CA 95322

NOTES:

PREPARED BY:

CARDOSO AG SERVICES INC.
PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: John Silveira Dairy #2

Physical address of dairy:

29687 Fentem RD	Gustine	Merced	95322
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1940

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X063-X080-X022-XXXX X063-X080-X023-XXXX

B. OPERATORS

Silveira, John

Operator name: Silveira, John

Telephone no.: Landline: ; Cellular: (209) 587-1796

P.O. Box 400	Gustine	CA	95322
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Silveira, John

Legal owner name: Silveira, John

Telephone no.: Landline: ; Cellular: (209) 587-1796

P.O. Box 400	Gustine	CA	95322
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	0	0	590	615	90	0
Maximum number	0	0	590	615	90	0
Average number	0	0	590	615	90	0
Avg live weight (lbs)	0	0	725	665		

Predominant milk cow breed: Jersey

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 11,698.83 *tons per reporting period*

Total nitrogen from manure: 118,953.50 *lbs per reporting period*

After ammonia losses (30% loss applied): 83,267.45 *lbs per reporting period*

Total phosphorus from manure: 19,677.52 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 28,979,998 *gallons*

Total nitrogen generated: 22,966.56 *lbs*

Total phosphorus generated: 5,825.84 *lbs*

Total potassium generated: 34,282.83 *lbs*

Total salt generated: 287,598.05 *lbs*

	28,979,998	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	28,979,998	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Ground water
Domestic Well	Ground water
Irrigation Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

Date	Material type / Description	Quantity	Reporting basis	Moist. (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	Separator solids *Solid Manure Imported from Dairy #1*	3,250.00 *ton*	Dry-weight	77.8	25,300.00	6,700.00	9,700.00		0.00

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	36,507.90	9,668.10	13,997.10	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	36,507.90	9,668.10	13,997.10	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	75	75	2	both	X063-X080-X023-XXXX
2	37	37	2	both	X063-X080-X022-XXXX
4	15	15	2	both	X063-X080-X022-XXXX
Totals for areas that were used for application	127	127	6		
Totals for areas that were not used for application					
Land application area totals	127	127	6		

B. CROPS AND HARVESTS

1

Field name: 1

11/09/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 75 Plant date: 11/09/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	1,227.80 *ton*	Dry-weight		68.1	18,200.00	3,300.00	21,700.00		9.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.37	190.09	34.47	226.65	981.78

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 75 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	2,494.50 *ton*	Dry-weight		69.8	13,500.00	2,500.00	15,000.00		5.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	33.26	271.20	50.22	301.34	1,165.16

2

Field name: 2

11/09/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 37 Plant date: 11/09/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	568.70 *ton*	Dry-weight		77.5	24,600.00	3,900.00	28,800.00		10.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.37	170.15	26.97	199.20	726.25

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	1,230.60 *ton*	Dry-weight		66.7	10,900.00	1,700.00	13,900.00		5.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	33.26	241.44	37.66	307.90	1,129.69

4

Field name: 4

11/09/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 15 Plant date: 11/09/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	229.20 *ton*	Dry-weight		74.2	26,400.00	4,300.00	30,100.00		12.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.28	208.15	33.90	237.32	1,001.33

4

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	498.90 *ton*	Dry-weight		69.6	13,400.00	2,100.00	15,000.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	33.26	270.98	42.47	303.33	1,152.66

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 11/09/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough Plant date: 11/09/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	139.61	53.44	86.84	0.00	1,988.00 *ton*
Application event totals		139.61	53.44	86.84	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	49.44	12.48	71.28	479.09	1,736,220.00 *gal*
Canal	Ground water	0.68	0.00	0.00	193.59	8,699,220.00 *gal*
Application event totals		50.11	12.48	71.28	672.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	68.53	15.48	102.22	739.62	1,984,252.00 *gal*
Canal	Ground water	0.86	0.00	0.00	246.69	11,085,732.00 *gal*
Application event totals		69.39	15.48	102.22	986.31	

1 - 06/01/2023: Corn, silage

Field name: 1

Crop: Corn, silage Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

1 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	1.42	0.00	0.00	405.16	18,206,776.00 *gal*
Application event totals		1.42	0.00	0.00	405.16	

05/08/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure Imported from Dairy #1	Separator solids	299.55	79.33	114.85	0.00	2,000.00 *ton*
Application event totals		299.55	79.33	114.85	0.00	

06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	24.51	6.09	39.41	341.11	2,976,378.00 *gal*
Canal	Ground water	1.42	0.00	0.00	405.16	18,206,776.00 *gal*
Application event totals		25.92	6.09	39.41	746.27	

07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	12.79	3.94	19.01	234.91	3,472,441.00 *gal*
Canal	Ground water	1.17	0.00	0.00	334.07	15,011,929.00 *gal*
Application event totals		13.96	3.94	19.01	568.98	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.72	0.00	0.00	205.58	9,238,110.00 *gal*
Application event totals		0.72	0.00	0.00	205.58	

1 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.70	4.22	20.37	251.69	3,720,472.00 *gal*
Canal	Ground water	0.78	0.00	0.00	222.28	9,988,707.00 *gal*
Application event totals		14.48	4.22	20.37	473.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	11.88	3.66	17.65	218.13	3,224,409.00 *gal*
Canal	Ground water	1.34	0.00	0.00	384.18	17,263,719.00 *gal*
Application event totals		13.22	3.66	17.65	602.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	1.05	0.00	0.00	298.95	13,433,752.00 *gal*
Application event totals		1.05	0.00	0.00	298.95	

2 - 11/09/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough

Plant date: 11/09/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	111.74	42.77	69.50	0.00	785.00 *ton*
Application event totals		111.74	42.77	69.50	0.00	

2 - 11/09/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	49.44	12.48	71.28	479.09	856,535.00 *gal*
Canal	Ground water	0.68	0.00	0.00	193.59	4,291,615.00 *gal*
Application event totals		50.11	12.48	71.28	672.68	

03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	68.53	15.48	102.22	739.62	978,898.00 *gal*
Canal	Ground water	0.86	0.00	0.00	246.69	5,468,961.00 *gal*
Application event totals		69.39	15.48	102.22	986.31	

2 - 06/01/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	1.42	0.00	0.00	405.16	8,982,009.00 *gal*
Application event totals		1.42	0.00	0.00	405.16	

05/08/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure Imported from Dairy #1	Separator solids	258.06	68.34	98.94	0.00	850.00 *ton*
Application event totals		258.06	68.34	98.94	0.00	

2 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	24.51	6.09	39.41	341.11	1,468,346.00 *gal*
Canal	Ground water	1.42	0.00	0.00	405.16	8,982,009.00 *gal*
Application event totals		25.92	6.09	39.41	746.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	12.79	3.94	19.01	234.91	1,713,071.00 *gal*
Canal	Ground water	1.17	0.00	0.00	334.07	7,405,885.00 *gal*
Application event totals		13.96	3.94	19.01	568.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.72	0.00	0.00	205.58	4,557,468.00 *gal*
Application event totals		0.72	0.00	0.00	205.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.70	4.22	20.37	251.69	1,835,433.00 *gal*
Canal	Ground water	0.78	0.00	0.00	222.28	4,927,762.00 *gal*
Application event totals		14.48	4.22	20.37	473.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	11.88	3.66	17.65	218.13	1,590,709.00 *gal*
Canal	Ground water	1.34	0.00	0.00	384.18	8,516,768.00 *gal*
Application event totals		13.22	3.66	17.65	602.31	

2 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	1.05	0.00	0.00	298.95	6,627,318.00 *gal*
Application event totals		1.05	0.00	0.00	298.95	

4 - 11/09/2022: Oats, silage-soft dough

Field name: 4

Crop: Oats, silage-soft dough

Plant date: 11/09/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	139.75	53.49	86.92	0.00	398.00 *ton*
Application event totals		139.75	53.49	86.92	0.00	

10/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	49.44	12.48	71.28	479.09	347,244.00 *gal*
Canal	Ground water	0.68	0.00	0.00	193.59	1,739,844.00 *gal*
Application event totals		50.11	12.48	71.28	672.68	

03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	68.53	15.48	102.22	739.62	396,850.00 *gal*
Canal	Ground water	0.86	0.00	0.00	246.69	2,217,146.00 *gal*
Application event totals		69.39	15.48	102.22	986.31	

4 - 06/01/2023: Corn, silage

4 - 06/01/2023: Corn, silage

Field name: 4

Crop: Corn, silage

Plant date: 06/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	1.42	0.00	0.00	405.16	3,641,355.00 *gal*
Application event totals		1.42	0.00	0.00	405.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure Imported from Dairy #1	Separator solids	299.55	79.33	114.85	0.00	400.00 *ton*
Application event totals		299.55	79.33	114.85	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	24.51	6.09	39.41	341.11	595,276.00 *gal*
Canal	Ground water	1.42	0.00	0.00	405.16	3,641,355.00 *gal*
Application event totals		25.92	6.09	39.41	746.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	12.79	3.94	19.01	234.91	694,488.00 *gal*
Canal	Ground water	1.17	0.00	0.00	334.07	3,002,386.00 *gal*
Application event totals		13.96	3.94	19.01	568.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.72	0.00	0.00	205.58	1,847,622.00 *gal*
Application event totals		0.72	0.00	0.00	205.58	

4 - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.70	4.22	20.37	251.69	744,094.00 *gal*
Canal	Ground water	0.78	0.00	0.00	222.28	1,997,741.00 *gal*
Application event totals		14.48	4.22	20.37	473.97	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	11.88	3.66	17.65	218.13	644,882.00 *gal*
Canal	Ground water	1.34	0.00	0.00	384.18	3,452,744.00 *gal*
Application event totals		13.22	3.66	17.65	602.31	

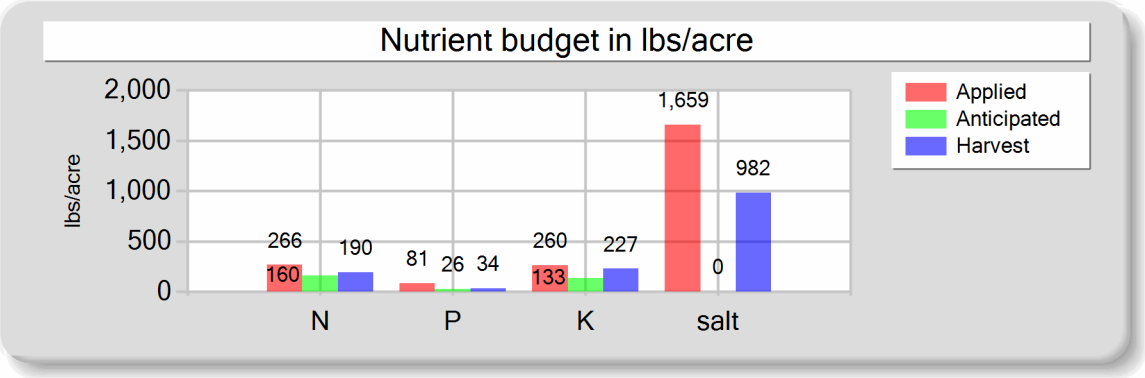
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	1.05	0.00	0.00	298.95	2,686,750.00 *gal*
Application event totals		1.05	0.00	0.00	298.95	

B. NUTRIENT BUDGET

1 - 11/09/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 11/09/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	139.61	53.44	86.84	0.00
Process wastewater	117.97	27.96	173.51	1,218.71
Fresh water	1.54	0.00	0.00	440.28
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	266.11	81.39	260.34	1,658.99
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	190.09	34.47	226.65	981.78
Nutrient balance	76.02	46.93	33.70	677.21
Applied to removed ratio	1.40	2.36	1.15	1.69

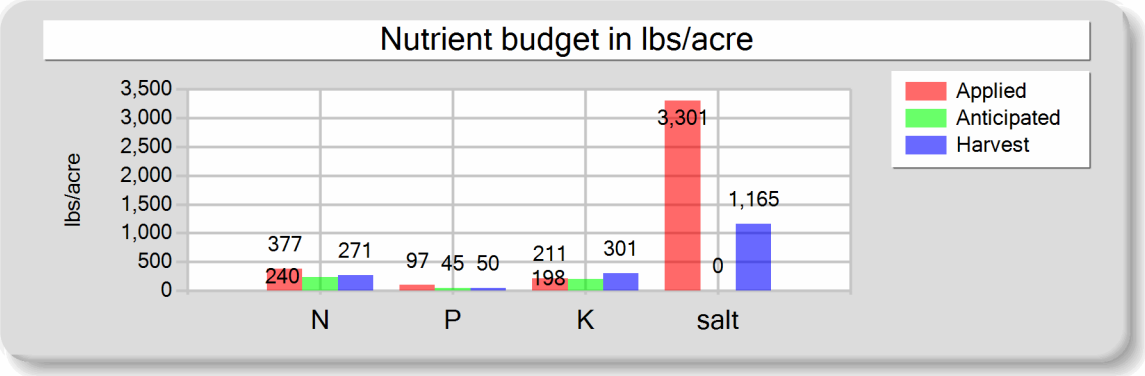
Fresh water applied	
19,784,952.00	*gallons*
728.61	*acre-inches*
9.71	*inches/acre*

Process wastewater applied	
3,720,472.00	*gallons*
137.01	*acre-inches*
1.83	*inches/acre*

Total harvests for the crop	
1	*harvests*

1 - 06/01/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	299.55	79.33	114.85	0.00
Process wastewater	62.87	17.92	96.44	1,045.84
Fresh water	7.89	0.00	0.00	2,255.37
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	377.32	97.24	211.29	3,301.21
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	271.20	50.22	301.34	1,165.16
Nutrient balance	106.12	47.02	-90.05	2,136.05
Applied to removed ratio	1.39	1.94	0.70	2.83

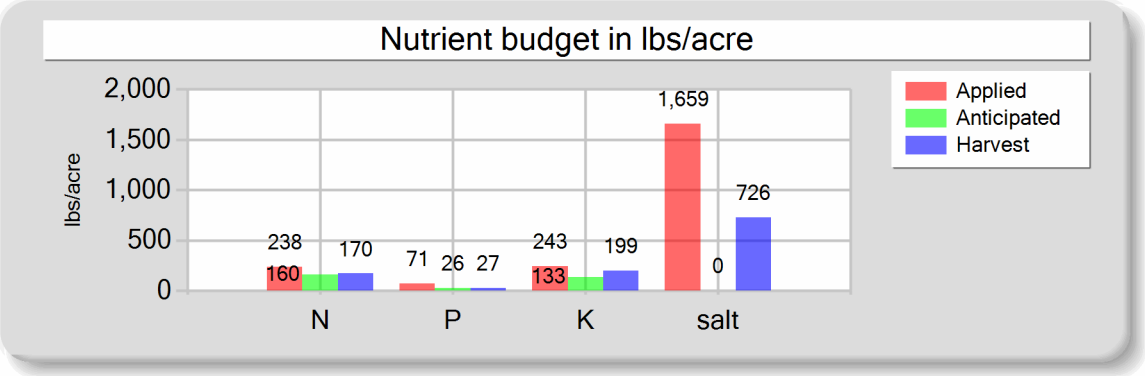
Fresh water applied	
101,349,769.00	*gallons*
3,732.37	*acre-inches*
49.76	*inches/acre*

Process wastewater applied	
13,393,700.00	*gallons*
493.24	*acre-inches*
6.58	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 11/09/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 11/09/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	111.74	42.77	69.50	0.00
Process wastewater	117.97	27.96	173.51	1,218.71
Fresh water	1.54	0.00	0.00	440.28
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	238.25	70.73	243.01	1,658.99
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	170.15	26.97	199.20	726.25
Nutrient balance	68.10	43.75	43.81	932.75
Applied to removed ratio	1.40	2.62	1.22	2.28

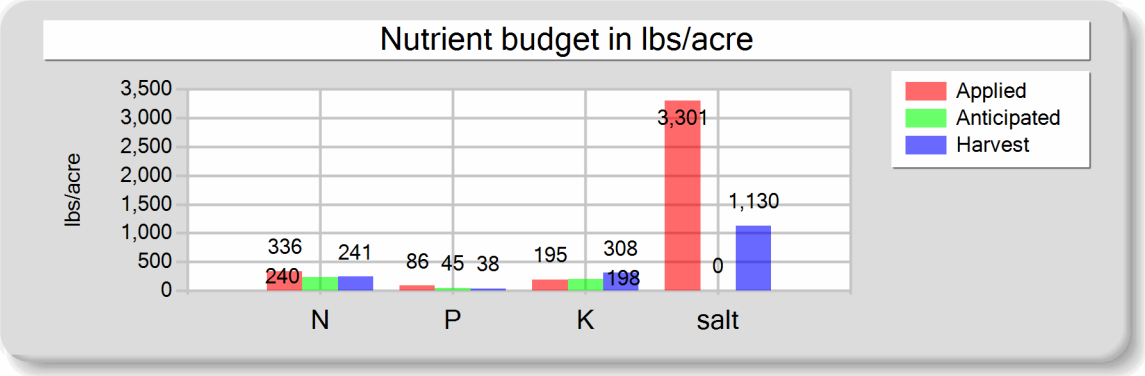
Fresh water applied	
9,760,576.00	*gallons*
359.45	*acre-inches*
9.71	*inches/acre*

Process wastewater applied	
1,835,433.00	*gallons*
67.59	*acre-inches*
1.83	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 06/01/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	258.06	68.34	98.94	0.00
Process wastewater	62.87	17.92	96.44	1,045.84
Fresh water	7.89	0.00	0.00	2,255.37
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	335.83	86.26	195.38	3,301.21
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	241.44	37.66	307.90	1,129.69
Nutrient balance	94.38	48.60	-112.52	2,171.52
Applied to removed ratio	1.39	2.29	0.63	2.92

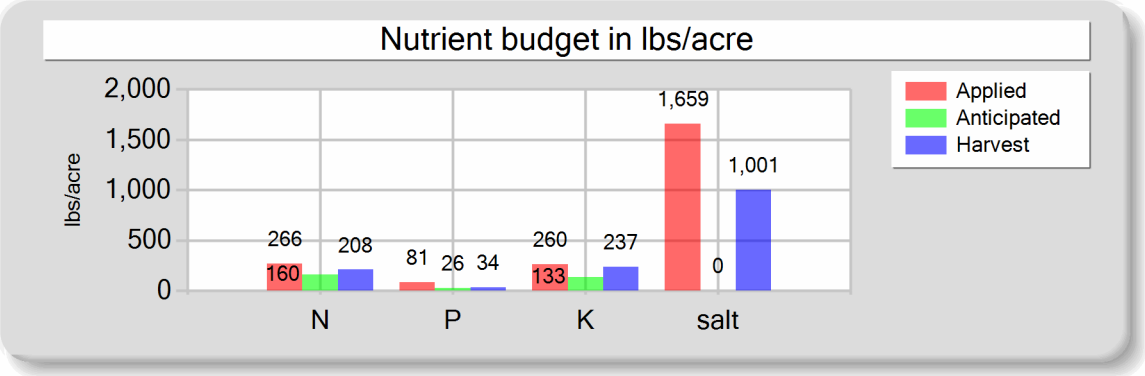
Fresh water applied	
49,999,219.00	*gallons*
1,841.30	*acre-inches*
49.76	*inches/acre*

Process wastewater applied	
6,607,559.00	*gallons*
243.33	*acre-inches*
6.58	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 11/09/2022: Oats, silage-soft dough

Field name: 4 Crop: Oats, silage-soft dough Plant date: 11/09/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	139.75	53.49	86.92	0.00
Process wastewater	117.97	27.96	173.51	1,218.71
Fresh water	1.54	0.00	0.00	440.28
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	266.25	81.45	260.43	1,658.99
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	208.15	33.90	237.32	1,001.33
Nutrient balance	58.10	47.54	23.11	657.66
Applied to removed ratio	1.28	2.40	1.10	1.66

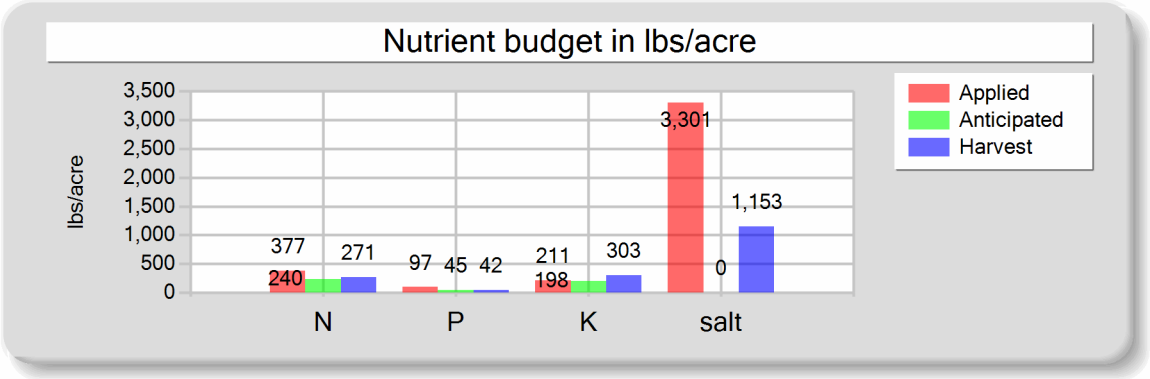
Fresh water applied	
3,956,990.00	*gallons*
145.72	*acre-inches*
9.71	*inches/acre*

Process wastewater applied	
744,094.00	*gallons*
27.40	*acre-inches*
1.83	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 06/01/2023: Corn, silage

Field name: 4 Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	299.55	79.33	114.85	0.00
Process wastewater	62.87	17.92	96.44	1,045.84
Fresh water	7.89	0.00	0.00	2,255.37
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	377.32	97.24	211.29	3,301.21
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	270.98	42.47	303.33	1,152.66
Nutrient balance	106.34	54.78	-92.05	2,148.55
Applied to removed ratio	1.39	2.29	0.70	2.86

Fresh water applied	
20,269,953.00	*gallons*
746.47	*acre-inches*
49.76	*inches/acre*

Process wastewater applied	
2,678,740.00	*gallons*
98.65	*acre-inches*
6.58	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Separator Manure - 2022

Sample and source description: Separator Manure - 2022

Sample date: 10/31/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 87.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,900.00	8,000.00	13,000.00							0.00
DL	100.00	100.00	100.00							1.00

Separator Manure

Sample and source description: Separator Manure

Sample date: 05/18/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 79.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,900.00	4,100.00	10,700.00							0.00
DL	0.01	0.01	0.01							1.00

Separator Manure

Sample and source description: Separator Manure

Sample date: 10/18/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 78.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,100.00	9,200.00	9,600.00							0.00
DL	0.01	0.01	0.01							1.00

B. PROCESS WASTEWATER ANALYSES

Lagoon Q4 - 2022

Sample and source description: Lagoon Q4 - 2022

Sample date: 11/22/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	255.00	196.00	0.00	0.90	64.60	369.00								4,950.00	2,480
DL	0.01	0.01	1.00	0.01	0.01	0.01								0.01	1

Lagoon Q1

Sample and source description: Lagoon Q1

Sample date: 03/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	309.00	226.00	0.00	1.40	70.10	463.00	182.00	92.70	146.00	1,880.00	0.00	50.10	394.00	5,220.00	3,350
DL	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	10.00	1

Lagoon Q2

Sample and source description: Lagoon Q2

Sample date: 05/18/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.20

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	72.80	53.10	0.00	1.20	18.40	119.00								1,980.00	1,030
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q3

Sample and source description: Lagoon Q3

Sample date: 07/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.10

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	32.90	22.20	0.00	0.20	10.20	49.20								932.00	608
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q4

Sample and source description: Lagoon Q4

Sample date: 11/06/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.10

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	90.50	45.10	0.00	2.40	18.00	112.00								2,440.00	1,520
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

C. FRESH WATER ANALYSES

Canal

CCID Canal

Sample description: CCID Canal

Sample date: 10/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	0.70								328.00	200
DL		0.01	0.01								0.01	1

Domestic Well

DW MB

Sample description: DW MB

Sample date: 12/14/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	17.70	0.00	17.70								1,730.00	0
DL	0.01	0.01	0.01								10.00	1

Irrigation Well

Irrigation Well

Irrigation Well

Sample description: Irrigation Well

Sample date: 11/06/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	15.50	0.00	15.50								1,730.00	0
DL	0.01	0.01	0.01								10.00	1

D. SOIL ANALYSES

1

Soil - Field 1

Sample and source description: Soil - Field 1

Sample date: 05/04/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			36.00				
DL			0.01				

2

Soil - Field 2

Sample and source description: Soil - Field 2

Sample date: 05/04/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			35.00				
DL			0.01				

4

4

Soil - Field 4

Sample and source description: Soil - Field 4

Sample date: 05/04/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			39.00				
DL			0.01				

E. PLANT TISSUE ANALYSES

1 - 11/09/2022: Oats, silage-soft dough

Field 1 Oat Silage

Sample and source description: Field 1 Oat Silage

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,200.00	3,300.00	21,700.00		9.40
DL	0.01	0.01	0.01		1.00

1 - 06/01/2023: Corn, silage

Field 1 Corn

Sample and source description: Field 1 Corn

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,500.00	2,500.00	15,000.00		5.80
DL	0.01	0.01	0.01		1.00

2 - 11/09/2022: Oats, silage-soft dough

Field 2 Oat Silage

Sample and source description: Field 2 Oat Silage

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,600.00	3,900.00	28,800.00		10.50
DL	0.01	0.01	0.01		1.00

2 - 06/01/2023: Corn, silage

Field 2 Corn

Sample and source description: Field 2 Corn

Sample date: 09/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,900.00	1,700.00	13,900.00		5.10
DL	0.01	0.01	0.01		1.00

4 - 11/09/2022: Oats, silage-soft dough

Field 4 Oat Silage

Sample and source description: Field 4 Oat Silage

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,400.00	4,300.00	30,100.00		12.70
DL	0.01	0.01	0.01		1.00

4 - 06/01/2023: Corn, silage

Field 4 Corn

Sample and source description: Field 4 Corn

Sample date: 09/14/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,400.00	2,100.00	15,000.00		5.70
DL	0.01	0.01	0.01		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

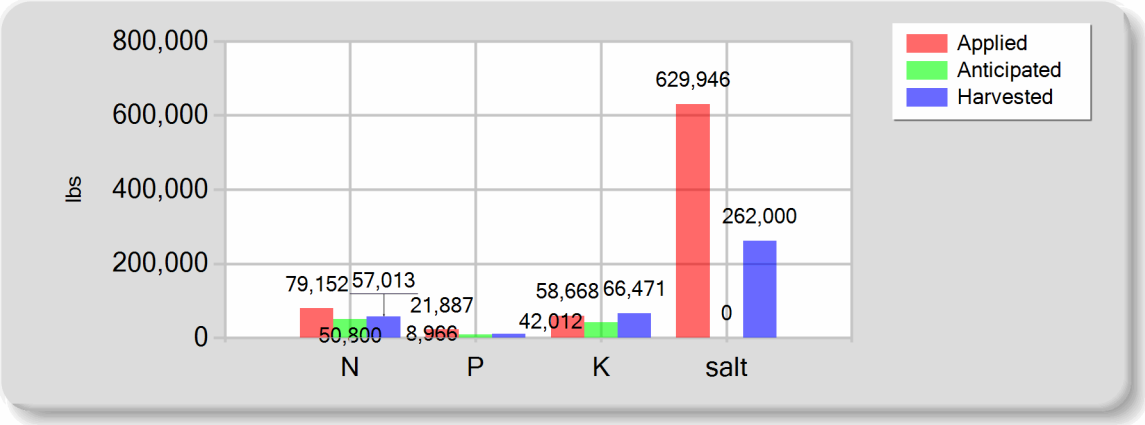
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

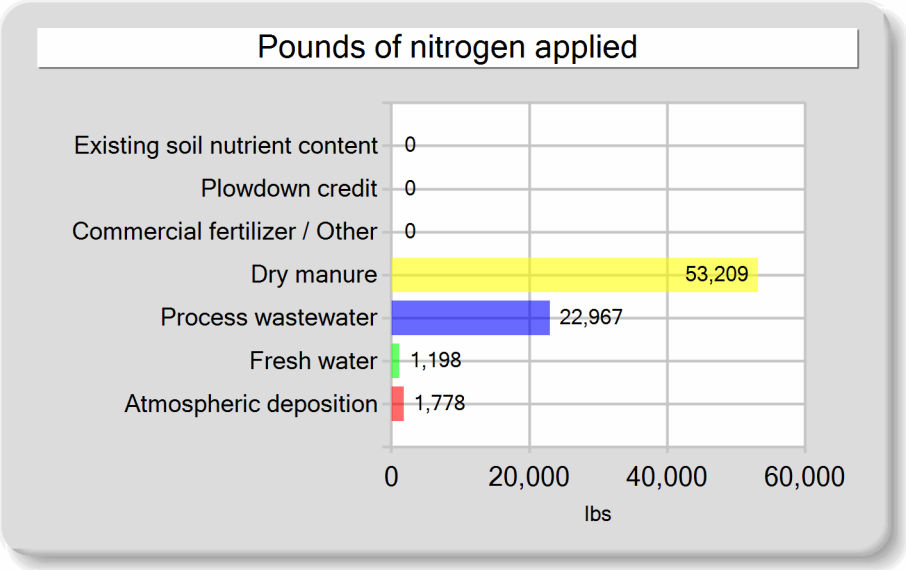
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

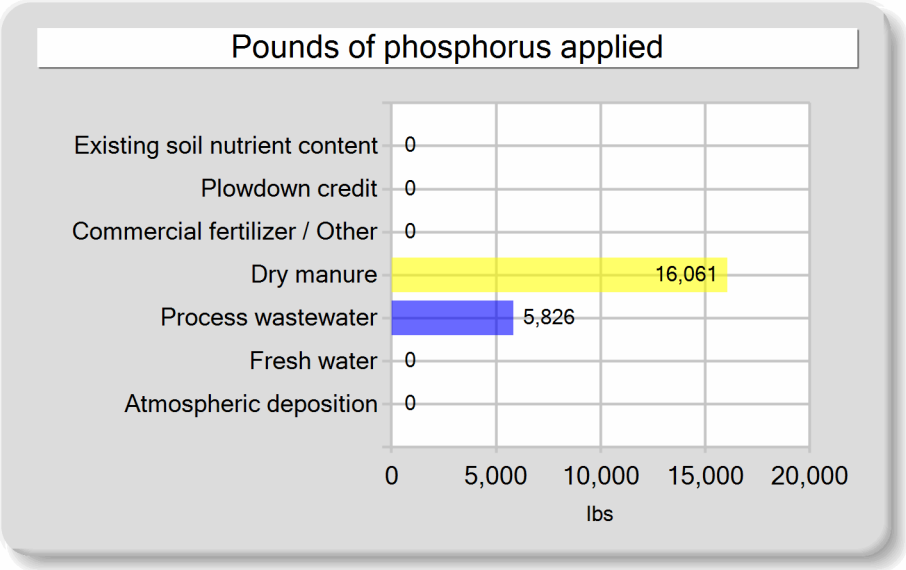
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	53,208.92	16,060.84	24,385.30	0.00
Process wastewater	22,966.56	5,825.84	34,282.83	287,598.05
Fresh water	1,198.22	0.00	0.00	342,347.72
Atmospheric deposition	1,778.00	0.00	0.00	0.00
Total nutrients applied	79,151.70	21,886.68	58,668.12	629,945.76
Anticipated crop nutrient removal	50,800.00	8,966.20	42,011.60	0.00
Actual crop nutrient removal	57,012.69	9,888.60	66,470.89	262,000.39
Nutrient balance	22,139.00	11,998.08	-7,802.77	367,945.37
Applied to removed ratio	1.39	2.21	0.88	2.40

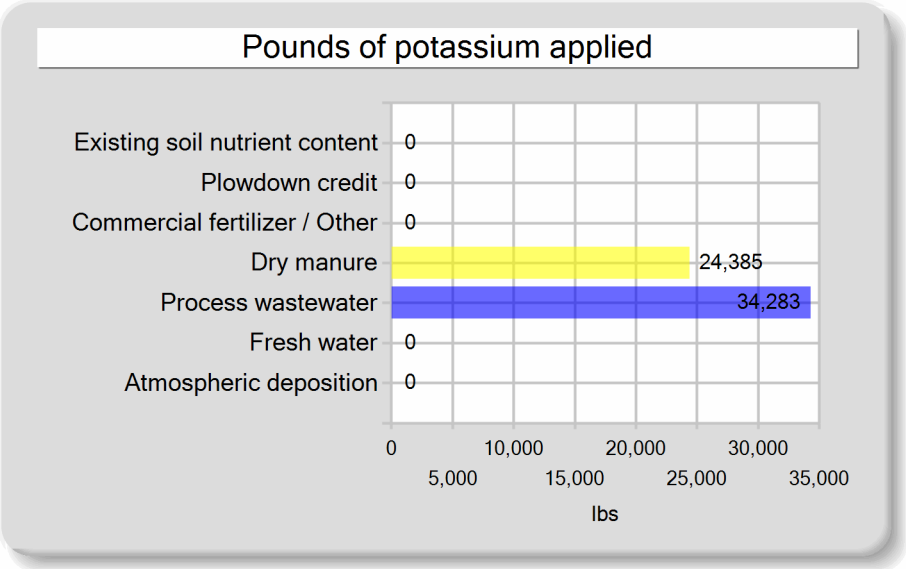
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

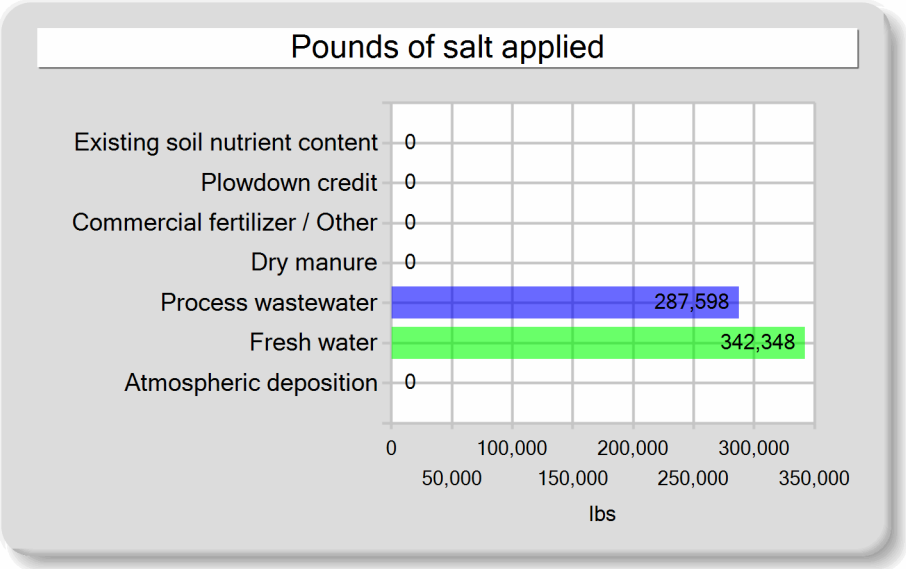


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

John Silveira

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
John Silveira	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 26, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Report of Dairy Well Water Analysis

DELLAVALLE LABORATORY INC

Lab No.: 23K0290
Sampled By: M Dekeyrel
Requested By: John Silveira
Submitted Date: 11/07/23
Reported Date: 11/14/23
Project:
Crop ID:

John Silveira Heifer Facility
PO Box 400
Gustine CA 95322
00-0023748 08

E-mail:
Copy To: cas.labs@yahoo.com

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	Irrigation Well	11/06/23	14:15	1730	1.73	15.5	ND		7.4

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

Report of Dairy Well Water Analysis

DELLAVALLE LABORATORY INC

John Silveira Heifer Facility
PO Box 400
Gustine CA 95322
00-0023748 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23K0290
Sampled By: M Dekeyrel
Requested By: John Silveira
Submitted Date: 11/07/23
Reported Date: 11/14/23
Project:
Crop ID:

		Date Sampled	Time Sampled	**EC** µmhos/cm	**EC** mmhos/cm	**NO_3-N** mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	Irrigation Well	11/06/23	14:15	**1730**	**1.73**	**15.5**	42.32	3.53

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

John Silveira Heifer Facility
PO Box 400
Gustine, CA 95322

Account# 00-0023748
Account Manager: Ben Nydam
Submitted By: John Silveira

Received: 11/07/2023 15:10
Reported: 11/14/2023 09:59

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23K0290-01	Irrigation Well	Ag Water	M Dekeyrel		11/06/2023 14:15

Default Cooler Temperature on Receipt °C: -0.8
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

John Silveira Heifer Facility
PO Box 400
Gustine, CA 95322

Account# 00-0023748
Account Manager: Ben Nydam
Submitted By: John Silveira

Received: 11/07/2023 15:10
Reported: 11/14/2023 09:59

Sample Results

Sample: Irrigation Well
23K0290-01 (Water)

Sampled: 11/6/2023 14:15
Sampled By: M Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.73**	mmhos/cm	0.01	1		11/10/23 14:51	SM 2510 B		BEK0299
Electrical Conductivity umhos	**1730**	umhos/cm	10.0	1		11/10/23 14:51	SM 2510 B		BEK0299
Ammonia (as N)	ND	mg/L	0.00	1		11/06/23 14:15	Field		BEK0178
Nitrate Nitrogen as NO3N	**15.5**	mg/L	0.1	1	10	11/07/23 20:13	EPA 300.0		BEK0232
Temperature	**25.0**	units	0.0	1		11/10/23 14:51	SM 4500-H+	H	BEK0299
pH	**7.4**	units	1.0	1		11/10/23 14:51	SM 4500-H+	H	BEK0299

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DELLAVALLE™
LABORATORY INC

John Silveira Heifer Facility
PO Box 400
Gustine, CA 95322

Account# 00-0023748
Account Manager: Ben Nydam
Submitted By: John Silveira

Received: 11/07/2023 15:10
Reported: 11/14/2023 09:59

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0232										
Blank (BEK0232-BLK1)					Prepared & Analyzed: 11/7/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0232-BLK2)					Prepared & Analyzed: 11/7/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEK0232-BS1)					Prepared & Analyzed: 11/7/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		104	90-110		
Duplicate (BEK0232-DUP1)		**Source: 23K0234-01**			Prepared: 11/7/2023 Analyzed: 11/8/2023					
Nitrate Nitrogen as NO3N	2.3		0.1	mg/L		2.3			1.21	10
Matrix Spike (BEK0232-MS1)		**Source: 23K0234-01**			Prepared & Analyzed: 11/7/2023					
Nitrate Nitrogen as NO3N	7.5		0.1	mg/L	5.000	2.3	104	90-110		
Reference (BEK0232-SRM1)					Prepared & Analyzed: 11/7/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEK0232-SRM2)					Prepared: 11/7/2023 Analyzed: 11/8/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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DELLAVALLE™
LABORATORY INC

John Silveira Heifer Facility
PO Box 400
Gustine, CA 95322

Account# 00-0023748
Account Manager: Ben Nydam
Submitted By: John Silveira

Received: 11/07/2023 15:10
Reported: 11/14/2023 09:59

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0299										
Blank (BEK0299-BLK1)					Prepared: 11/9/2023	Analyzed: 11/10/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
pH	6.0		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEK0299-BLK2)					Prepared: 11/9/2023	Analyzed: 11/10/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.7		1.0	units						
Duplicate (BEK0299-DUP1)		**Source: 23K0350-02**			Prepared: 11/9/2023	Analyzed: 11/10/2023				
Electrical Conductivity	1.38		0.01	mmhos/cm					200	10
pH	5.0		1.0	units		5.0			0.00	10
Electrical Conductivity umhos	1380		10.0	umhos/cm					200	10
Reference (BEK0299-SRM1)					Prepared: 11/9/2023	Analyzed: 11/10/2023				
Electrical Conductivity	570			umhos/cm	538.0		106	90-110		
Reference (BEK0299-SRM2)					Prepared: 11/9/2023	Analyzed: 11/10/2023				
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEK0299-SRM3)					Prepared: 11/9/2023	Analyzed: 11/10/2023				
Electrical Conductivity	1090			umhos/cm	1000		109	90-110		
Electrical Conductivity umhos	1090			umhos/cm	1000		109	90-110		
Reference (BEK0299-SRM4)					Prepared: 11/9/2023	Analyzed: 11/10/2023				
Electrical Conductivity	1090			umhos/cm	1000		109	90-110		
Electrical Conductivity umhos	1090			umhos/cm	1000		109	90-110		
Reference (BEK0299-SRM6)					Prepared: 11/9/2023	Analyzed: 11/10/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0299-SRM7)					Prepared: 11/9/2023	Analyzed: 11/10/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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11/07/23 15:30 23K0290

WATER WORK REQUEST

Acct No. Cons.

Bill To: 23748 8

Purchase Order No. Results Needed By

Client **John Silveira Heifer Facility**
Address PO Box 400
City, State, Zip Gustine, CA 95322
Phone Fax
Cell/Email

Copy to Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by John Silveira - (209) 678-2797

Ranch

Date sampled 11/6/23

Sampled by mDekeuvel

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

No. of Samples 1 No. Bottles

Water Type: [ ] Drinking [ ]Wastewater
[ ] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other

Analysis and Bottles Required: *(Please Indicate Analysis)*

(✓) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test)**
(1) 1 L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) 1 L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) 1 L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) 1 L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) 1 L plastic, unpreserved (white)

( ) Other

DESCRIPTION OF SAMPLES		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. Irrigation well	Sampled From:	11/6/23	2150	Ø	-0.8
2.	Sampled From:				
3.	Sampled From:				
4.	Sampled From:				
5.	Sampled From:				
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	mDekeuvel	CASIbc	11/6/23 2150	11/7/23 1400
Second	JLF	DLV	11-7-23 1:40	11-7-23 2:52
Third				
Fourth	[illegible]	DLI	11/7 15:10	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling Hrs ___ Miles ___ Consulting ___

Amt Paid Rec By Check No. Date

Shipping

$ ___ In
$ ___ Out

Signature

Sample received in cooler with ice?
[ ] Yes [ ] No

ctt:update 2020

Shipping Information: Shipped In ☐ Picked-Up ☑ Walk In ☐ DLI Sampler ☐ Other ☐

Container: Ice Chest ☑ Box ☐ None ☐ **Refrigerant:** Wet Ice ☑ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

11/07/23 15:30 23K0290

	Type of Container(s) Received	Sample Number 1	2	3	4	5	6	7	8	9	10
	Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
	Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

SEND OUT

Report of Dairy Well Water Analysis

DELLAVALLE LABORATORY INC

John Silveira Heifer Facility
PO Box 400
Gustine CA 95322
00-0023748 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L1017
Sampled By: M.Pedroso
Requested By: John Silveira
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	DW MB	12/14/23	9:35	1730	1.73	17.7	ND		7.8

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

John Silveira Heifer Facility
PO Box 400
Gustine CA 95322
00-0023748 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L1017
Sampled By: M.Pedroso
Requested By: John Silveira
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	**EC** µmhos/cm	**EC** mmhos/cm	**NO_3-N** mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	DW MB	12/14/23	9:35	**1730**	**1.73**	**17.7**	48.32	4.03

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

John Silveira Heifer Facility
PO Box 400
Gustine, CA 95322

Account# 00-0023748
Account Manager: Ben Nydam
Submitted By: John Silveira

Received: 12/14/2023 14:20
Reported: 12/21/2023 10:27

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L1017-01	DW MB	Ag Water	M.Pedroso		12/14/2023 9:35

Default Cooler Temperature on Receipt °C: 7.3
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Sample Results

Sample: DW MB
23L1017-01 (Water)

Sampled: 12/14/2023 9:35
Sampled By: M.Pedroso

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.73**	mmhos/cm	0.01	1		12/19/23 12:02	SM 2510 B		BEL0706
Electrical Conductivity umhos	**1730**	umhos/cm	10.0	1		12/19/23 12:02	SM 2510 B		BEL0706
Ammonia (as N)	ND	mg/L	0.00	1		12/14/23 09:35	Field		BEL0657
Nitrate Nitrogen as NO3N	**17.7**	mg/L	0.1	1	10	12/19/23 00:40	EPA 300.0		BEL0686
Temperature	**25.0**	units	0.0	1		12/19/23 12:02	SM 4500-H+	H	BEL0706
pH	**7.8**	units	1.0	1		12/19/23 12:02	SM 4500-H+	H	BEL0706

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Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0686										
Blank (BEL0686-BLK1)					Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0686-BLK2)					Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0686-BLK3)					Prepared: 12/18/2023 Analyzed: 12/19/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEL0686-BS1)					Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		101	90-110		
LCS (BEL0686-BS2)					Prepared: 12/18/2023 Analyzed: 12/19/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		102	90-110		
Duplicate (BEL0686-DUP1)		**Source: 23L1009-01**			Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L		5.1			0.974	10
Duplicate (BEL0686-DUP2)		**Source: 23L1009-08**			Prepared: 12/18/2023 Analyzed: 12/19/2023					
Nitrate Nitrogen as NO3N	5.3		0.1	mg/L		5.3			0.530	10
Matrix Spike (BEL0686-MS1)		**Source: 23L1009-01**			Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	10.4		0.1	mg/L	5.000	5.1	106	90-110		
Matrix Spike (BEL0686-MS2)		**Source: 23L1009-08**			Prepared: 12/18/2023 Analyzed: 12/19/2023					
Nitrate Nitrogen as NO3N	10.4		0.1	mg/L	5.000	5.3	102	90-110		
Reference (BEL0686-SRM1)					Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEL0686-SRM2)					Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEL0686-SRM3)					Prepared: 12/18/2023 Analyzed: 12/19/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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Reported: 12/21/2023 10:27

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0706										
Blank (BEL0706-BLK1)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	5.6		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEL0706-BLK2)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.9		1.0	units						
Blank (BEL0706-BLK3)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
pH	6.8		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEL0706-DUP1)		**Source: 23L1017-01**			Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	1.74		0.01	mmhos/cm		1.73			0.403	10
pH	7.7		1.0	units		7.8			0.258	10
Electrical Conductivity umhos	1740		10.0	umhos/cm		1730			0.403	10
Duplicate (BEL0706-DUP2)		**Source: 23L1088-01**			Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	0.52		0.01	mmhos/cm		0.52			0.116	10
Electrical Conductivity umhos	517		10.0	umhos/cm		517			0.116	10
pH	6.7		1.0	units		6.8			0.297	10
Reference (BEL0706-SRM1)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	440			umhos/cm	426.0		103	90-110		
Reference (BEL0706-SRM2)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0706-SRM3)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	1040			umhos/cm	1000		104	90-110		
Electrical Conductivity umhos	1040			umhos/cm	1000		104	90-110		
Reference (BEL0706-SRM4)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEL0706-SRM5)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0706 (Continued)										
Reference (BEL0706-SRM5)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEL0706-SRM6)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0706-SRM7)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0706-SRM8)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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12/14/23 14:20 23L1017

WATER WORK REQUEST

Acct No. Cons.

Bill To: 23748 8

Purchase Order No. Results Needed By

Client: John Silveira Heifer Facility
Address: PO Box 400
City, State, Zip: Gustine, CA 95322
Phone: Fax:
Cell/Email:

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: John Silveira - (209) 678-2797

Ranch:

Date sampled: 12/14/23

Sampled by: mPedroso

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

No. of Samples: 1 No. Bottles:

Water Type: [ ] Drinking [ ] Wastewater
[X] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other

Analysis and Bottles Required: *(Please Indicate Analysis)*

(X) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test)**
(1) l L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) l L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) l L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) l L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) l L plastic, unpreserved (white)

( ) Other

DESCRIPTION OF SAMPLES		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. DW MB	Sampled From:	12/14/23	935A	0	7.3
2.	Sampled From:				
3.	Sampled From:				
4.	Sampled From:				
5.	Sampled From:				
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	mPed	CASInc	12/14/23 935A	12/14/23 12pm
Second	CSM	DLI	12/14/23 12pm	12/14/23 2:15pm
Third				
Fourth	[illegible]	DLI	12/14 14:00	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling Hrs ___ Miles ___ Consulting ___

Amt Paid ___ Rec By ___ Check No. ___

Shipping
$ ___ In
$ ___ Out
Date ___

Signature

Sample received in cooler with ice?
[ ] Yes [ ] No

ctl:update 2020

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

Container: Ice Chest ☒ Box ☐ None ☐ **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

12/14/23 14:20 23L1017

Type of Container(s) Received		Sample Number 1	2	3	4	5	6	7	8	9	10
Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

SEND OUT